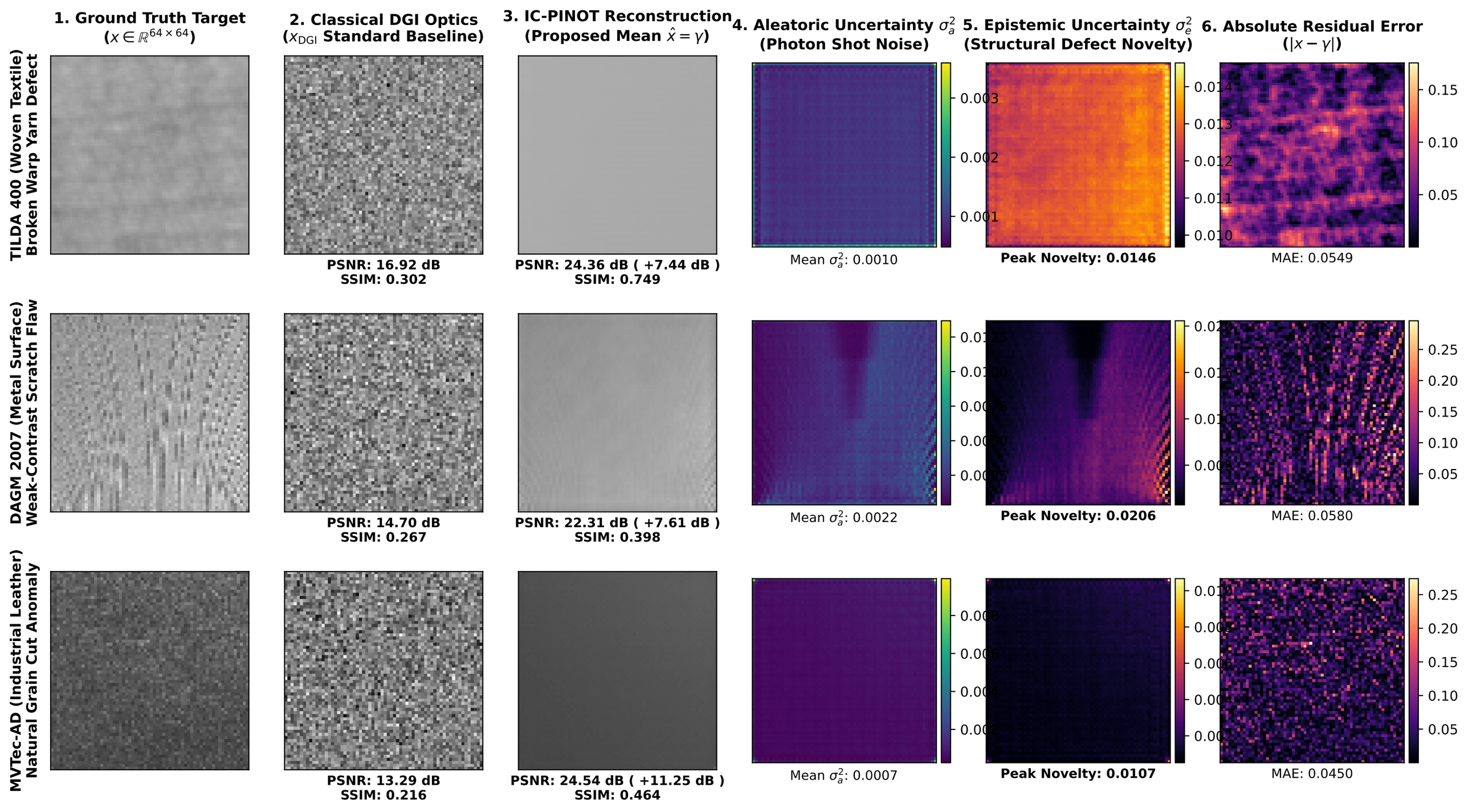

1. Ground Truth Target
($x \in \mathbb{R}^{64 \times 64}$)
2. Classical DGI Optics
(x_{DGI} Standard Baseline)
3. IC-PINOT Reconstruction
(Proposed Mean $\hat{x} = \gamma$)
4. Aleatoric Uncertainty σ_a^2
(Photon Shot Noise)
5. Epistemic Uncertainty σ_e^2
(Structural Defect Novelty)
6. Absolute Residual Error
($|x - \gamma|$)
TILDA 400 (Woven Textile)
Broken Warp Yarn Defect
PSNR: 16.92 dB
SSIM: 0.302
PSNR: 24.36 dB (+7.44 dB)
SSIM: 0.749
Mean σ_a^2: 0.0010
Peak Novelty: 0.0146
MAE: 0.0549
DAGM 2007 (Metal Surface)
Weak-Contrast Scratch Flaw
PSNR: 14.70 dB
SSIM: 0.267
PSNR: 22.31 dB (+7.61 dB)
SSIM: 0.398
Mean σ_a^2: 0.0022
Peak Novelty: 0.0206
MAE: 0.0580
MVTec-AD (Industrial Leather)
Natural Grain Cut Anomaly
PSNR: 13.29 dB
SSIM: 0.216
PSNR: 24.54 dB (+11.25 dB)
SSIM: 0.464
Mean σ_a^2: 0.0007
Peak Novelty: 0.0107
MAE: 0.0450